MissionForge

Short Project Report | Astroverse

Shashank Pandey · Individual project · 7 November 2026

1. Problem and proposed solution

Spaceflight concepts are difficult to understand from equations or launch animations alone. A rocket can reach space and still return to Earth because altitude alone does not provide the sideways velocity required for orbit. Students and space enthusiasts need a way to connect design choices, physical principles and observable consequences.

MissionForge is a browser-based spacecraft builder, flight simulator and learning platform. Learners predict an outcome, inspect or build a rocket, fly it, compare the result with a reference calculation and explain their next change. The design hypothesis is that this cycle can make orbital concepts easier to understand; measured educational effectiveness remains future work.

2. Core experience and implemented features

Area	Implemented experience
Engineering	36 construction parts, seven presets, staging, fuel and delta-v calculations, readiness checks and three difficulty modes.
Campaign	Four missions: First Orbit, Earth Observer, Lunar Orbiter, and Crewed Lunar Flyby and Return. Reports expose objectives, events and telemetry.
Learning	Five Academy lessons, an optional first-flight guide, NASA Data Explorer, interactive debriefs, a reflection journal and model comparison.
Exploration	Eleven destinations, three stock craft, maps, maneuvers and local fleets. A restricted adapter imports compatible single-stage Hangar builds.

3. Problem-solving approach and innovation

The distinctive contribution is the connection between **traceable source data, an executable model and a prediction–flight–reflection activity**. NASA values affect calculations that the learner can inspect. Incorrect predictions are retained, and debriefs relate outcomes to available evidence without inventing missing telemetry or treating event proximity as proof of causation.

The dependable showcase is First Orbit on Amateur using Orbiter-9: inspect readiness, launch with guidance, observe sideways acceleration, check periapsis and explain the result. Prepared checkpoints can shorten a demonstration but must be identified. Longer English and Hindi walkthroughs are available in the repository.

Data, science and validation

A visible chain from original source fields to a simulation result.

4. NASA data and provenance

Eight pinned snapshots from NASA’s official GMAT repository comprise five data files and three method references. The planetary kernel is JPL/NAIF pck00011. Original bytes, notices and SHA-256 hashes are retained; derived SI-unit data is generated separately. Retrieval was on 4 November 2026 from commit **c9b78bd1c8499845cb78fde315e1f761b6b06f1c**. Archived GMAT defaults are not presented as current precision measurements. [1–3]

Source input	Use in MissionForge
NAIF radii and rotation fields	12 spherical radii and 12 signed linear rotation periods; body geometry, altitude and simplified spin.
GMAT gravitational parameters	10 source parameters for gravitational and orbital calculations; other assumed values remain labelled.
Earth / Mars atmosphere tables	28 Earth and 150 Mars density bands for documented drag paths. These are archived models, not live weather.
EGM96 and method references	Optional degree/order-8 Earth harmonics; density, transfer and burn references support separately implemented methods.

5. Science made useful

Source axes in kilometres become a volume-equivalent spherical radius: **$R = 1000 \times (a \times b \times c)^{(1/3)}$**. Gravitational parameters convert by **$\mu[m^3/s^2] = \mu[km^3/s^2] \times 10^9$**. For a circular two-body orbit, **$v = \sqrt{(\mu/(R+h))}$**. At 200 km, the reference speed is approximately **7.788 km/s on Earth** and **3.454 km/s on Mars**. Changing the selected world therefore changes a meaningful flight reference.

The ideal rocket equation connects mass, propellant and delta-v. Numerical integration propagates motion; atmospheric density, drag and finite burns change the flown result. Paired model comparison starts from the same state to illustrate how selected forces alter trajectories. Its difference also includes numerical error, so it is not a direct measure of physical truth.

6. Recorded validation evidence

Recorded check	Scope and interpretation
321 regressions; 621,913 generated checks	Separate finite test scopes in the recorded 30-group release audit of 6 November 2026.
12 campaign outcomes; 330 device/world/craft cases	Four missions × three difficulties, including an expected budget-limited partial result; device profiles are emulated.
Independent numerical references	60-digit Decimal fixtures test Earth/Moon/Mars circular speed, escape speed and period at 10^{-10} relative tolerance.
Follow-up, 7 November 2026	Recorded mobile-layout checks include 14 layouts and 33 small-phone world/craft starts. [4,5]

Evidence is taken from checked-in audit records; this document did not rerun the full audit. Numerical agreement supports the tested reference conditions, not every trajectory, real spacecraft or physical device.

Implementation and next steps

Accessible delivery, explicit boundaries and a repeatable demonstration.

7. Architecture and accessibility

React provides the interface and Three.js/WebGL renders the worlds and spacecraft. Local JavaScript simulation and workers support physics, planning and scientific experiments. Scientific inputs are bundled. IndexedDB supports persistent local data, while exports provide portable records. Prepared service-worker caching enables offline play after an initial connected setup.

Desktop controls include WASD steering and remappable keys. Touch controls expose steering, throttle, staging and RCS, with the camera area separated from the control deck and secondary readings in a drawer. Themes, responsive layouts and optional guidance support different users. Static hosting delivers the application; gameplay needs no hosted simulator, multiplayer service, account or runtime scientific API key.

8. Implementation boundaries and limitations

Campaign and exploration use different documented force paths. Approximate planetary positions, simplified materials and sandbox placements must not be described as professional mission navigation or simulated transfers. Build-to-Exploration supports a restricted compatible class, not every possible rocket. Advanced laboratories include 342-constituent station loading, convex-mesh fracture and a QuickJS WebAssembly interpreter backend; these are not a global ocean-height atlas, calibrated arbitrary material fracture or a native C/Rust engine port.

Recorded browser emulation does not certify real Android/iOS hardware or every browser. A formal learner study has not yet established learning gains. The reported live URL and reviewed walkthroughs are access points; exact hosted-build identity and offline recovery should be checked before presenting. NASA-derived data use does not imply NASA endorsement.

9. Presentation plan and next milestone

Three-minute showcase: introduce the altitude-versus-orbit misconception; compare sourced Earth/Mars values; inspect Orbiter-9; show a prepared First Orbit flight; inspect periapsis and the debrief; finish with the learner’s explanation. Keep the English/Hindi recordings and a prepared offline build as backups.

Next evidence: run a small consent-based learner trial with a before/after explanation task; test the hosted origin on physical phone and desktop hardware; verify saves and prepared offline startup. Prioritise these checks before expanding feature count. Success should mean that a learner can explain a result using data and flight evidence, not merely complete a lesson.

References and access

Live application: mission-forge.pandeyshashankai.workers.dev
Project repository: github.com/pandeyshashankai/Mission-Forge

[1] NASA GMAT — pinned source repository

[2] JPL/NAIF — pck00011 planetary constants

[3] Project data provenance and scientific scope

[4] Recorded comprehensive validation, 6 Nov 2026

[5] Mobile-layout follow-up, 7 Nov 2026

[6] Connected learning and model/adapter boundaries